Assignment 1: Question Formation and Exploratory Analysis

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Draft Questions

I am currently taking the Research Project A course, and as such my question forming process has been following closely alongside it. I started my research with a focus on robot vision and an interest in space research. The first Big Question I arrived at was, "*How can I advance domain adaptation for space robotics vision?",* however it was pointed out to me, by my supervisor, how broad robot vision was. Our discussions on refining my scope led to the topic of *traversability estimation* and how domain adaptation would still play a part. With the scope of this class in mind, I thought of a possible draft question branching from the basis of my research topic. That question being, "*Can I combine various moon maps/photos to find where a rover could safely drive and where it might get stuck?".* Answering this question can help create safer rover routes. This would help save money and lower the risk during new Moon landing missions.

Datasets

- Geosciences Node Lunar Orbital Data Explorer (ODE) of NASA's Planetary Data System (PDS)
    - Lunar Reconnaissance Orbiter (LRO)
        - Lunar Orbiter Laser Altimeter (LOLA)
        - Lunar Reconnaissance Orbiter Camera (LROC)
            - Engineering Data Records (EDR)
        - Miniature Radio Frequency (Mini-RF)
- Astropedia – Lunar and Planetary Cartographic Catalog from NASA's PDS
    - Moon Apollo Traverse Maps

The datasets I managed to find consist of the LOLA height map, Narrow Angle Camera (NAC) photos, radar images, and old rover tracks. These all fit the description of Big Data, as was described in Module 1. These datasets are large, contain different data types, come from several different sources, and contain real-world labels.

Some of the datasets seem to have inconsistencies like photos with lighting streaks, which can be solved with image preprocessing, or examples of "stuck rovers", which we can create new labels for based on the slope.

Refined & Backup Questions

In order to work around possible inconsistencies in the data, to improve model accuracy, and to reduce resources/time required, I thought it would be beneficial to narrow down our scope to 1 area. This makes a new potential question: *"Inside a 10x10 degree patch near the lunar south pole, which mix of slope, surface texture, and radar signal is best able to flag non-traversable terrain?"* Alternatively, I could explore if, *"Inside a 50x50 km region near the lunar south pole, does adding high-resolution surface texture to a basic slope map improve our ability to flag non-traversable terrain?"*

As a back-up we can research, *"Is radar alone able to determine whether terrain is risky or not?",* and we would simply just use the Mini-RF radar map as a backup dataset.

References

LOLA Science Team 2014, *Moon LRO LOLA DEM 118 m* (GDRDEM v2), dataset, USGS Astrogeology Science Center, Flagstaff, viewed 15 June 2025, https://astrogeology.usgs.gov/search/map/moon_lro_lola_dem_118m.

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NASA Planetary Data System 2014, *LRO Mini-RF Global Mosaic Data Set v1.0* (data set ID LRO-L-MRFLRO-5-GLOBAL-MOSAIC-V1.0), dataset, PDS Geosciences Node, Washington University in St Louis, viewed 15 June 2025, https://pds.nasa.gov/ds-view/pds/viewDataset.jsp?dsid=LRO-L-MRFLRO-5-GLOBAL-MOSAIC-V1.0.

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